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Curvature of an unstable manifold

Surface geometry and the lower Rössler equilibrium

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Curvature of a curve in Euclidean space

For a curve $\gamma(u)$ with arclength s and speed $v = \|\gamma'(u)\|$. The **unit tangent** is

$$\boldsymbol{t} = \frac{d\gamma}{ds} = \frac{\gamma'(u)}{v}.$$

The **curvature** is

$$\kappa = \left\| \frac{d\boldsymbol{t}}{ds} \right\| = \frac{\|\gamma'(u) \times \gamma''(u)\|}{v^3} \geq 0.$$

Frenet frame in $\mathbb{R}^3$ with $\kappa > 0$.

The derivative $d\boldsymbol{t}/ds$ is perpendicular to $\boldsymbol{t}$ because $\boldsymbol{t} \cdot \boldsymbol{t} = 1$.

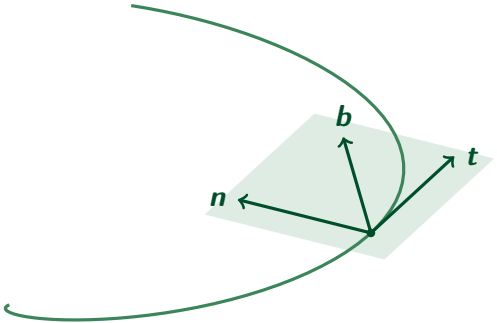
$$\boldsymbol{t} = \frac{\boldsymbol{\gamma}'(u)}{v} \qquad \text{unit tangent,}$$

$$\boldsymbol{n} = \frac{1}{\kappa}\frac{d\boldsymbol{t}}{ds} \qquad \text{principal normal,}$$

$$\boldsymbol{b} = \boldsymbol{t} \times \boldsymbol{n} = \frac{\boldsymbol{\gamma}'(u) \times \boldsymbol{\gamma}''(u)}{\|\boldsymbol{\gamma}'(u) \times \boldsymbol{\gamma}''(u)\|} \qquad \text{binormal.}$$

The speed v normalizes $\boldsymbol{\gamma}'(u)$. The curvature κ normalizes $d\boldsymbol{t}/ds$. The cross-product magnitude is $v^3\kappa$. The cross product $\boldsymbol{b}$ automatically has length 1.

Frenet frame for a helix



The frame is shown at $u = 0$ on $\gamma(u) = (\cos u, \sin u, u/3)$. The shaded **osculating plane** contains $\boldsymbol{t}$ and $\boldsymbol{n}$. The binormal $\boldsymbol{b}$ is perpendicular to it.

Torsion and the Frenet–Serret equations

With the arclength s parametrization

$$\boxed{\boldsymbol{b}.\boldsymbol{b} = 1 \quad \text{and} \quad \boldsymbol{b} \cdot \boldsymbol{t} = 0 \quad \text{and} \quad \boldsymbol{t}' = \kappa \boldsymbol{n}} \Longrightarrow \boxed{\boldsymbol{b}' \cdot \boldsymbol{b} = \boldsymbol{b}' \cdot \boldsymbol{t} = 0}$$

Thus $\boldsymbol{b}' = -\tau \boldsymbol{n}$ with $\tau = -\boldsymbol{b}' \cdot \boldsymbol{n}$ and $\boldsymbol{n} = \boldsymbol{b} \times \boldsymbol{t}$ gives $\boldsymbol{n}' = -\kappa \boldsymbol{t} + \tau \boldsymbol{b}$. Together,

$$\boxed{\boldsymbol{t}' = \kappa \boldsymbol{n}, \qquad \boldsymbol{n}' = -\kappa \boldsymbol{t} + \tau \boldsymbol{b}, \qquad \boldsymbol{b}' = -\tau \boldsymbol{n}.}$$

Curvature measures how fast the tangent turns. Torsion measures how fast the osculating plane rotates about the tangent.

XXXX Normal curvature and principal curvatures

For a curve on a surface, let $\boldsymbol{n}$ now denote the unit surface normal. The normal curvature in the direction of the unit tangent $\boldsymbol{t}$ is

$$\kappa_n(\boldsymbol{t}) = \langle d\boldsymbol{t}/ds, \boldsymbol{n} \rangle.$$

Its maximum and minimum over unit tangent directions are the **principal curvatures** k_1, k_2.

$$K = k_1 k_2 \quad \text{(Gaussian)}, \qquad H = \tfrac{1}{2}(k_1 + k_2) \quad \text{(mean)}.$$

Reversing $\boldsymbol{n}$ reverses the signs of k_1, k_2, H. The product K keeps its value.

A chart with orthonormal tangent directions

At $\boldsymbol{p}_0$, choose orthonormal tangent directions $\boldsymbol{e}_1, \boldsymbol{e}_2$ and set $\boldsymbol{n} = \boldsymbol{e}_1 \times \boldsymbol{e}_2$. Write the surface locally as

$$\boldsymbol{p}(u_1, u_2) = \boldsymbol{p}_0 + u_1\boldsymbol{e}_1 + u_2\boldsymbol{e}_2 + h(u_1, u_2)\boldsymbol{n}.$$

Here $h(0) = 0$ and $Dh(0) = 0$. For a unit tangent $\boldsymbol{t} = v_1\boldsymbol{e}_1 + v_2\boldsymbol{e}_2$, write $\boldsymbol{v} = (v_1, v_2)^{\mathsf{T}}$. At $\boldsymbol{p}_0$,

$$B = D^2h(0), \qquad \kappa_n(\boldsymbol{t}) = \boldsymbol{v}^{\mathsf{T}} B \boldsymbol{v}.$$

The eigenvalues of B are k_1, k_2, so $K = \det B$ and $H = \frac{1}{2}\operatorname{tr} B$.

Curvature of a graph over the coordinate plane

For $\boldsymbol{p}(u,v) = (u, v, h(u,v))$, set $\boldsymbol{g} = \nabla h$, $Q = D^2 h$, and $r = \sqrt{1 + \|\boldsymbol{g}\|^2}$. The graph directions $\boldsymbol{p}_i = \partial_i \boldsymbol{p}$ have dot products $G_{ij} = \langle \boldsymbol{p}_i, \boldsymbol{p}_j \rangle$.

$$\boldsymbol{n} = \frac{(-\boldsymbol{g}, 1)}{r}, \qquad G = I + \boldsymbol{g}\boldsymbol{g}^{\mathsf{T}}, \qquad B = \frac{Q}{r}.$$

The principal curvatures solve $\det(B - kG) = 0$, and

$$K = \frac{\det B}{\det G} = \frac{\det Q}{(1 + \|\boldsymbol{g}\|^2)^2}.$$

Example at the origin

For $h(u,v) = \frac{1}{2}(u^2 - v^2)$, $G = I$ at the origin. Then $k_1 = 1,\ k_2 = -1,\ K = -1,\ H = 0$.

The unstable surface as an invariant graph

Near a hyperbolic equilibrium, suppose the two-dimensional unstable eigenspace projects invertibly onto the x_1y_1-plane. Set $\boldsymbol{w} = (x_1, y_1)^{\mathsf{T}}$ in shifted coordinates. The local unstable manifold is $z_1 = h(\boldsymbol{w})$.

$$\frac{d\boldsymbol{w}}{dt} = \boldsymbol{f}(\boldsymbol{w}, z_1), \qquad \dot{z}_1 = j(\boldsymbol{w}, z_1).$$

The chain rule and tangency of the vector field give

$$Dh(\boldsymbol{w})\boldsymbol{f}(\boldsymbol{w}, h(\boldsymbol{w})) = j(\boldsymbol{w}, h(\boldsymbol{w})).$$

Expand h at the equilibrium and match powers of $\boldsymbol{w}$.

$$h(\boldsymbol{w}) = \boldsymbol{g}^{\mathsf{T}}\boldsymbol{w} + \tfrac{1}{2}\boldsymbol{w}^{\mathsf{T}}Q\boldsymbol{w} + O(\|\boldsymbol{w}\|^3).$$

The unstable eigenspace supplies $\boldsymbol{g}$. The quadratic terms determine the symmetric matrix Q, which gives the curvature at the equilibrium.

The lower Rössler equilibrium

Use the original Euclidean state coordinates (x, y, z) and the parameters $a = b = 0.2,\ c = 5.7$.

$$\dot{x} = -y - z, \qquad \dot{y} = x + ay, \qquad \dot{z} = b + z(x - c).$$

Solving the equilibrium equations gives

$$z_- = \frac{c - \sqrt{c^2 - 4ab}}{2a}, \qquad \boldsymbol{p}_- = (az_-, -z_-, z_-) \approx (0.0070262, -0.0351310, 0.0351310).$$

With $d = az_- - c$, the Jacobian and its eigenvalues are

$$J = \begin{pmatrix} 0 & -1 & -1 \\ 1 & a & 0 \\ z_- & 0 & d \end{pmatrix}, \qquad \begin{aligned} \lambda_s &\approx -5.686976, \\ \lambda_{u,\pm} &\approx 0.097001 \pm 0.995193\, i. \end{aligned}$$

The complex pair spans a real two-dimensional unstable plane.

The tangent plane from the stable left eigenvector

Shift the equilibrium to the origin with $(x_1, y_1, z_1) = (x, y, z) - \boldsymbol{p}_-$. The shifted system is

$$\dot{x}_1 = -y_1 - z_1, \qquad \dot{y}_1 = x_1 + a y_1, \qquad \dot{z}_1 = z_- x_1 + d z_1 + x_1 z_1.$$

A left eigenvector $\boldsymbol{\ell}$ for λ_s annihilates the two unstable eigenvectors and is normal to the unstable plane. Scale it to have third component 1.

$$\boldsymbol{\ell} = (-p, -q, 1)^{\mathsf{T}}, \qquad \boldsymbol{g} = \begin{pmatrix} p \\ q \end{pmatrix} \approx \begin{pmatrix} 0.00599829 \\ 0.00101891 \end{pmatrix}.$$

On the plane $z_1 = p x_1 + q y_1$, the linearized dynamics have matrix

$$A = \begin{pmatrix} -p & -1 - q \\ 1 & a \end{pmatrix}, \qquad \lambda_s = d + p.$$

The Hessian from a three-variable linear solve

The invariance equation for the shifted graph is

$$h_{x_1}(-y_1 - h) + h_{y_1}(x_1 + a y_1) = z_- x_1 + d h + x_1 h.$$

With $\boldsymbol{e}_1 = (1, 0)^{\mathsf{T}}$, matching quadratic terms gives

$$\boldsymbol{w}^{\mathsf{T}} Q A \boldsymbol{w} - \tfrac{\lambda_s}{2} \boldsymbol{w}^{\mathsf{T}} Q \boldsymbol{w} = (\boldsymbol{e}_1^{\mathsf{T}} \boldsymbol{w})(\boldsymbol{g}^{\mathsf{T}} \boldsymbol{w}).$$

Taking the symmetric matrix of this quadratic identity yields

$$\boxed{A^{\mathsf{T}} Q + Q A - \lambda_s Q = \boldsymbol{e}_1 \boldsymbol{g}^{\mathsf{T}} + \boldsymbol{g} \boldsymbol{e}_1^{\mathsf{T}}}.$$

The 11, 12, and 22 entries are three linear equations for Q_{11}, Q_{12}, Q_{22}. The eigenvalues of A have positive real part, so their pairwise sums cannot equal $\lambda_s < 0$.

Curvatures at the equilibrium

Solving for the Hessian of the local unstable graph gives

$$Q \approx \begin{pmatrix} 1.945588 \times 10^{-3} & 4.777028 \times 10^{-4} \\ 4.777028 \times 10^{-4} & 1.571189 \times 10^{-4} \end{pmatrix}.$$

Solving $\det(B - kG) = 0$ with the upward normal gives

$$k_1 \approx 2.06507 \times 10^{-3}, \quad k_2 \approx 3.75207 \times 10^{-5},$$
$$H \approx 1.05130 \times 10^{-3}, \quad K \approx 7.74830 \times 10^{-8}.$$

The surface bends toward the upward normal in both principal directions. The saddle-focus label describes the equilibrium's flow.

Curvature at nearby points

The quadratic graph gives the exact derivatives through order two at the equilibrium, up to numerical roundoff in the coefficients.

For a nearby point, match higher powers in the invariance equation to compute a higher-degree polynomial h_m. Evaluate ∇h_m and $D^2 h_m$ at the point and use the graph formulas.

$$r_m(\boldsymbol{w}) = Dh_m(\boldsymbol{w})\boldsymbol{f}(\boldsymbol{w}, h_m(\boldsymbol{w})) - j(\boldsymbol{w}, h_m(\boldsymbol{w})).$$

A degree-m Taylor graph has residual $O(\|\boldsymbol{w}\|^{m+1})$ near the origin. Compare the curvature at successive degrees, since the curvature calculation differentiates the graph twice.

References

O. E. Rössler, *An Equation for Continuous Chaos*, Physics Letters A 57 (1976), 397–398. Article

X. Cabré, E. Fontich, and R. de la Llave, *The Parameterization Method for Invariant Manifolds I. Manifolds Associated to Non-resonant Subspaces*, Indiana University Mathematics Journal 52 (2003), 283–328. Paper